SPECTRALFORGE: A PHYSICS-BASED SIMULATOR FOR FLUORESCENCE HYPERSPECTRAL IMAGING

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ABSTRACT

Acquiring four-dimensional fluorescence hyperspectral data is slow and costly. An instrument records a full emission spectrum at each of several excitation wavelengths and for every pixel, fluorescence signals are weak and easily buried in scatter and autofluorescence, and the illumination needed to raise them photobleaches the sample, so a single cube can consume hours of scarce instrument time. This makes it hard to plan acquisitions, prototype processing methods, or build shared benchmarks. We present **SpectralForge**, an open, physics-based simulator that renders realistic 4D excitation-emission cubes with perfect ground truth from a parametric fluorophore model obeying the standard laws of fluorescence photophysics (Kasha, Vavilov, Franck-Condon, Stokes shift), with optional inner-filter attenuation, reabsorption, Foerster resonance energy transfer (FRET), scatter, photon and read noise, and instrument clutter. Because every rendered cube comes with its concentration maps and true emission peaks, a method can be developed, tested, and compared before any measurement is taken. As one worked example, we use the simulator to study spectral band selection, where the goal is to keep only the informative bands and so acquire or process less. Scenes are built so that band variance is decoupled from class informativeness (measured correlation ≈ -0.1), so the task is not solved by variance ranking, and two controls, a random baseline and the full spectrum, tell a real gain from a chance one. Running a panel of selectors through the simulator shows it behaving as designed and marks where label-free selection helps and where it does not. SpectralForge and its workbench are in active development, with early access available to interested labs on request; the full study is released for reproducible use in 4D fluorescence imaging.

***Index Terms*—** fluorescence hyperspectral imaging, physics-based simulation, synthetic data, excitation-emission, photophysics, band selection.

1. INTRODUCTION

A multi-excitation fluorescence measurement illuminates a sample at several excitation wavelengths and, for each, records an emission spectrum across many bands. Every pixel therefore carries an excitation-emission matrix (EEM), and an image is a four-dimensional array over excitation, emission, height, and width (Fig. 1). The modality is useful because different fluorophores, tissue states, or species separate in this joint excitation-emission domain, which is why it is used across biomedical and material imaging, for example label-free imaging of cardiac tissue by its endogenous autofluorescence [9, 10]. Adding excitation wavelengths turns each single spectrum into an excitation-emission matrix that separates overlapping fluorophores a single-excitation hyperspectral image cannot, which improves downstream tissue and material classification [12–14]. It is also expensive to acquire. Each added excitation multiplies the acquisition by a full emission scan, fluorescence signals are weak and easily lost in scatter and autofluorescence, and the illumination needed to raise them photobleaches the very molecules being measured. A single high-quality 4D cube can therefore consume hours of specialized instrument time, and labeled real cubes remain scarce.

This scarcity slows everything downstream. Planning an acquisition, prototyping a processing method, or comparing algorithms all ideally happen before instrument time is committed, yet each needs representative data, and real cubes carry no ground truth: there is no reference for which fluorophores are where or which bands truly matter. A physically realistic simulator addresses both problems at once. If it renders cubes from the actual laws of fluorescence and returns the underlying concentrations and emission peaks, a method can be built, tested, and benchmarked against an answer key, and only the promising design need ever reach the instrument.

This paper presents **SpectralForge**, an open simulator of that kind for 4D fluorescence imaging, with an interactive workbench for building scenes by hand and a small library for scripting them. To show the tool at work we take one concrete task, spectral band selection, and run a study through it end to end: the simulator provides the cubes and their ground truth, a controlled evaluation protocol scores the candidate selectors, and the results show both that the tool behaves as designed and where a learned selector actually helps. Band selection is only one use. The same generator supports experiment planning, algorithm prototyping, teaching, and any study that benefits from 4D fluorescence data with a known answer.

2. THE SPECTRALFORGE FORWARD MODEL

2.1. Photophysics

The generator is built on the physics of fluorescence so that "informative" means what it means physically. A fluorophore absorbs a photon, relaxes non-radiatively to the lowest vibrational level of the first excited singlet, and emits. Four consequences shape the EEM. By *Kasha's rule* emission occurs from that lowest level regardless of the excitation, so the emission shape is independent of excitation wavelength and the EEM is approximately trilinear (an excitation

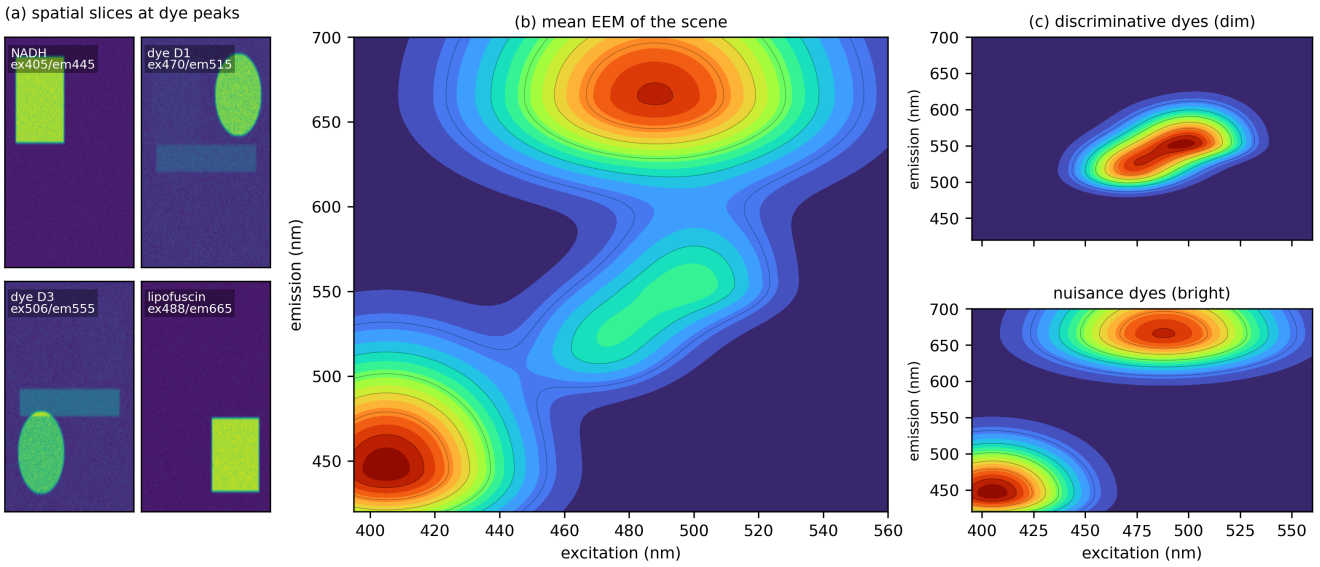


Fig. 1. A rendered 4D fluorescence cube from SpectralForge, here an example scene painted in the workbench with each dye laid down as a shape. (a) A slice of the cube taken at each dye's emission peak reveals that dye's painted region (NADH square, dyes D1/D3 disks, lipofuscin block); a slice lights up only dyes that both absorb at its excitation and emit at its emission, so every pixel effectively stores an excitation-emission matrix (EEM). (b) The mean EEM of the scene is the familiar fluorescence landscape: each fluorophore is a smooth blob at its excitation and (Stokes-shifted) emission. (c) Decomposed by role, the discriminative dyes form a single dim, overlapping cluster while the class-irrelevant nuisance dyes are bright and well separated, so band variance is dominated by the uninformative signal.

profile times an emission profile times a concentration map), the basis of PARAFAC. By *Vavilov's rule* the quantum yield is, to first order, independent of excitation; only the amount absorbed changes. The *Franck-Condon* principle sets the broadened, roughly mirror-image band envelopes and an optional vibronic shoulder about $1300\,\mathrm{cm}^{-1}$ to the red. The *Stokes shift* red-shifts emission relative to absorption.

2.2. Trilinear render and the additive invariant

Each fluorophore k is a parametric model with a peak-normalized Gaussian excitation profile $a_k(\lambda_{\mathrm{ex}})$ and an area-normalized emission profile $e_k(\lambda_{\mathrm{em}})$, extinction ε_k, quantum yield Φ_k, and a per-pixel concentration map $c_k(x, y)$. The rendered cube is a dilute linear superposition,

$$F(x, y, \lambda_{\mathrm{ex}}, \lambda_{\mathrm{em}}) = g \sum_k c_k(x, y)\, \varepsilon_k\, \Phi_k\, a_k(\lambda_{\mathrm{ex}})\, e_k(\lambda_{\mathrm{em}}), \quad (1)$$

where g folds lamp, exposure, and power. This is Kasha's rule written as algebra. The profiles are Gaussian, a_k peak-normalized and e_k area-normalized with widths $\mathrm{FWHM}/(2\sqrt{2\ln 2})$, and a fluorophore's brightness is the product $\varepsilon_k\Phi_k$. With all optional physics and artifacts off the model is *exactly linear*, $\mathrm{render}(A{+}B) = \mathrm{render}(A){+}\mathrm{render}(B)$, an invariant the implementation tests; difficulty is then added in realistic, opt-in increments. A per-pixel class label is the argmax over the discriminative fluorophores. Fig. 2 shows the fluorophore model.

2.3. Variance is decoupled from informativeness

The central design choice is the scene generator. Discriminative dyes are painted with smooth random concentration fields whose per-pixel argmax defines the class; they are dim and spectrally overlapping, so their content lives in low-variance band *shape*, while independent nuisance dyes are bright but class-irrelevant (Fig. 1c). Real photophysics compounds the decoupling: scatter ridges and bright autofluorescence add high-variance but class-irrelevant signal, shot noise inflates bright redundant bands, and the discriminative signal sits in dim or subtly shaped bands. On a confounded scene the correlation between per-band variance and per-band discriminability is ≈ -0.1, versus strongly positive on naive clean data, so the highest-variance bands carry no class information (Fig. 3) and a variance ranker cannot solve the task.

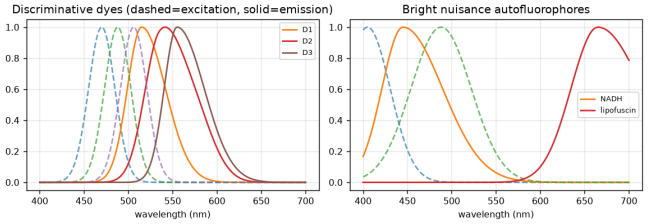


Fig. 2. Parametric fluorophore spectra. Left: three discriminative dyes (dashed, excitation; solid, emission) of modest brightness with overlapping emission and distinct excitation. Right: two bright nuisance autofluorophores that dominate band variance but carry no class information.

2.4. Artifacts, nonlinear physics, and clutter

On top of the clean cube the generator adds Rayleigh scatter at $\lambda_{\mathrm{em}} = \lambda_{\mathrm{ex}}$ (with a second-order ghost at $2\lambda_{\mathrm{ex}}$) and a water Raman

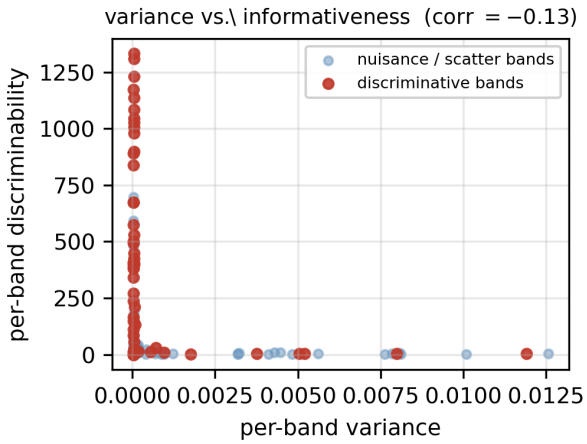


Fig. 3. Variance is decoupled from informativeness. Each point is a band: per-band variance against per-band discriminability. The high-variance bands are nuisance and scatter bands, not the discriminative ones (correlation near zero), so ranking bands by variance fails.

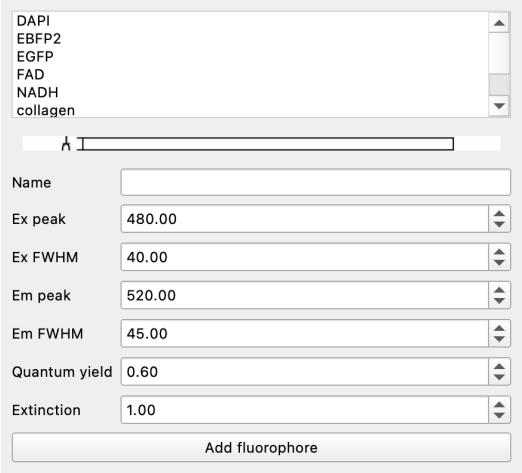


Fig. 4. Defining a fluorophore in the workbench library: choose a built-in dye or enter excitation and emission peaks and widths, quantum yield, and extinction. Materials are concentration mixtures of these fluorophores.

line at a fixed wavenumber offset ($\approx 3400\,\mathrm{cm}^{-1}$), each scaled by a per-pixel turbidity field, then Poisson shot noise and Gaussian read noise. Three optional effects introduce concentration nonlinearity: a primary inner filter $\exp(-sA_{\mathrm{ex}})$; *reabsorption* $\exp(-sA_{\mathrm{em}}(\lambda))$, whose overlap with the blue edge of emission reshapes the band and relocates discriminative information into band shape; and FRET, where a saturating fraction $E = kc_A/(1 + kc_A)$ of donor energy is transferred to a co-localized acceptor, encoding dye co-localization in a band ratio. Compactly, the inner filter and transfer are

$$F = F_{\mathrm{obs}}\, 10^{-(A_{\mathrm{ex}}+A_{\mathrm{em}})/2}, \qquad E = \frac{k\, c_A}{1 + k\, c_A}, \tag{2}$$

with reabsorption $A_{\mathrm{em}}(\lambda) = \sum_k \varepsilon_k c_k\, \alpha_k(\lambda)$ and a Raman line at $1/\lambda_R = 1/\lambda_{\mathrm{ex}} - \Delta\bar{\nu}$. Finally, fixed-pattern clutter injects many class-irrelevant low-rank spatial fields on random band subsets, the analogue of illumination and detector artifacts and the dominant real-instrument confound. By dialing these knobs the generator spans *clean* (variance ≈ informativeness, an idealized baseline that real acquisitions never meet), *realistic*, *clutter*, *nonlinear* (reabsorption or FRET), and *spatial-texture* regimes from a single call.

2.5. Ground truth, reproducibility, and real fluorophores

Every render returns a ground-truth sidecar with concentration maps, clean cubes, and per-fluorophore emission peaks, which enables a tight *peak-recovery* metric (did a selected band land within a tolerance of a true emission peak; a random selector scores about 0.33) in addition to broad information-retention metrics. The generator exposes a one-call `build_dataset` API with explicit knobs for size, emission sampling, nuisance brightness, scatter, noise, clutter, and nonlinear physics; a script re-runs every simulation and records exact parameters, so figures and numbers are reproducible. Parametric dyes can be replaced by real measured spectra imported from FPbase, and an interactive workbench lets a user paint concentration maps, render, and validate a selection against ground truth. This makes SpectralForge usable both as a controlled benchmark and as a design tool for real 4D fluorescence experiments. As an interactive application (Fig. 5), a user picks or defines fluorophores in a library (Fig. 4: excitation and emission peaks and widths, quantum yield, extinction), mixes them into materials that are concentration recipes, paints those materials onto a canvas as layers, then acquires, renders the 4D cube, selects bands, and validates against ground truth, all in one window.

3. EVALUATION PROTOCOL

Selection quality is the accuracy of a downstream classifier trained on the selected bands, using labels only at evaluation time. We score each subset with a classifier panel (k-nearest neighbors, random forest, MLP, logistic regression), cross-validated, reporting macro-F1 and, on nonlinear regimes, a *nonlinear-only gap* (the accuracy an expressive model recovers beyond a linear one), because a linear metric cannot credit a nonlinear-shape selection. Two controls matter here. A **random** baseline: a method that cannot beat the 95th percentile of random band subsets adds nothing. And the **full spectrum**: the accuracy ceiling a selector tries to approach. Where discrimination is spatial we use spatial-block cross-validation to remove spatial-autocorrelation leakage. These two controls, often absent in prior work, guard against reporting a gain that a random subset would already reach.

4. USING THE BENCHMARK

Band selection, which removes non-informative bands to shrink a real hyperspectral acquisition [11, 15], is one thing SpectralForge is built to test, so we exercise it with a panel of standard selectors: variance ranking, PCA loadings, supervised mutual information (a labels-using reference), and a fully unsupervised autoencoder-with-latent-perturbation selector, which reconstructs the standardized cube, perturbs each latent dimension, and turns the per-band reconstruction sensitivity into a per-band influence. The generator supplies the cubes and their ground truth; the protocol above scores each subset. The question is not which selector wins but whether the benchmark behaves as designed, that is, whether decoupling variance from informativeness makes it discriminate methods and

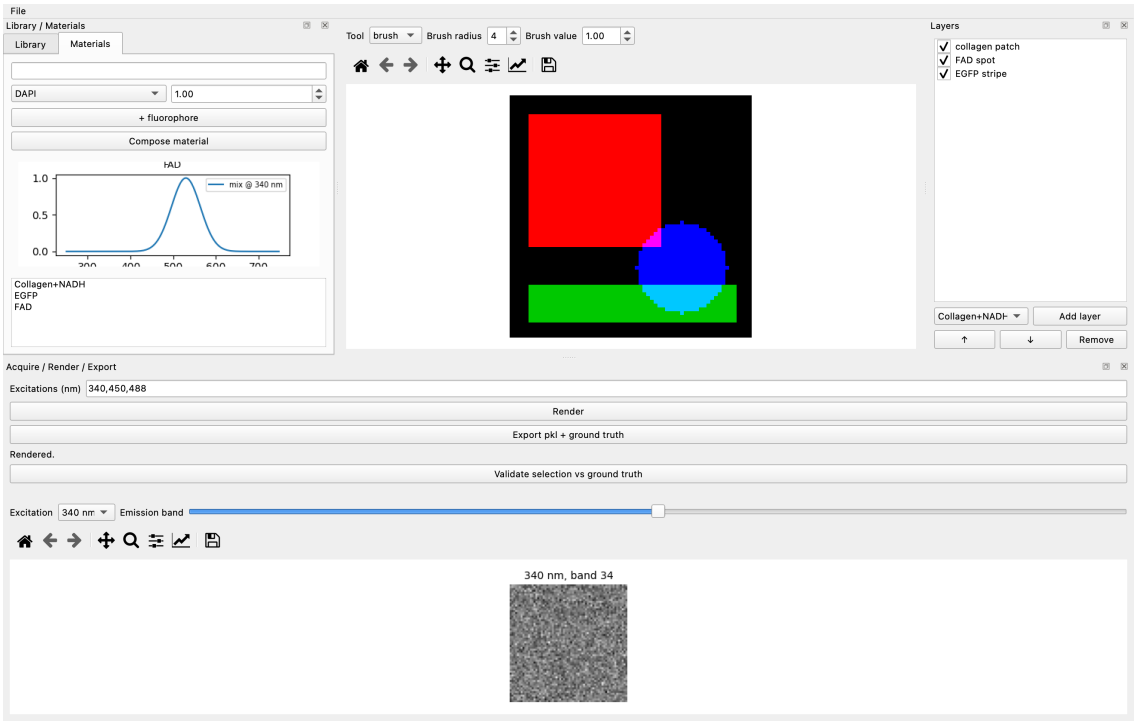


Fig. 5. The SpectralForge workbench. Left: fluorophore recipes (materials) with a live spectrum. Centre: the painted canvas (a collagen and NADH patch, an FAD spot, an EGFP stripe), coloured by class label. Right: the layer stack. Bottom: acquire, render, and one-click validation against ground truth, here previewing a rendered emission band at 340 nm.

Table 1. Label-free autoencoder vs. the strongest blind baseline (PCA loadings): best-of-panel macro-F1 over six seeds. Bold marks the better selector; the clean and reabsorption margins are significant at 95%, the rest are ties.

Regime	AE	PCA-load	verdict
clean	0.590	**0.635**	linear suffices
realistic	0.527	0.531	tie
reabsorption	**0.449**	0.431	label-free wins
FRET	0.635	0.642	tie

regimes.

It does. On confounded scenes a variance ranker barely beats random. PCA loadings is the strongest blind selector only in the idealized *clean* regime, where the informative bands are also the brightest. No real fluorescence acquisition reaches that regime, since real cubes always carry scatter, autofluorescence, and noise; once those enter, on the realistic and FRET regimes, the two selectors tie. Label-free selection overtakes PCA only where the physics places discriminative information in nonlinear band *shape*, as under reabsorption (macro-F1 0.449 vs 0.431, 95% CI $[+0.001, +0.029]$, Cohen $d \approx +0.94$; its subset carries nonlinear-only information $+0.030$ that the linear selector's does not, -0.021; Table 1). The pattern is a general one: *informative implies has variance implies PCA-selectable*, so a nonlinear *classification* task need not imply a nonlinear *selection* task, and the benchmark separates the two. This regime map, not a single winner, is what the benchmark reports.

The random and full-spectrum controls also mark the limits (Fig. 6). Under fixed-pattern clutter no blind selector beats a random subset, because clutter is high-variance but class-irrelevant; there only label-aware selection helps. No selector beats the full spectrum, so selection buys compression, not accuracy, and blind subsets are unstable band to band (Jaccard about 0.1 to 0.2), which favors selecting spectral regions or consensus sets. And when class is carried by spatial texture, so that every class shares the same per-pixel spectrum, per-pixel selection falls below random (0.52 to 0.55, against a random 0.674) while a spatial model with the right bands reaches the oracle (0.78). These are properties the benchmark makes visible, and they answer the practical questions: when to select bands, when to keep the spectrum, and when the problem is spatial rather than spectral.

5. DISCUSSION AND CONCLUSION

SpectralForge gives 4D fluorescence imaging an open, physically realistic source of cubes with perfect ground truth, so that methods can be designed, tested, and compared before scarce instrument time is spent. The band-selection study is one demonstration of that value. Paired with a protocol whose random and full-spectrum controls check that a reported gain is real, the simulator turns "does this selector help" into a measured, reproducible question, and its answer is specific: PCA loadings is the strongest blind baseline only in the idealized clean regime that no real acquisition reaches, ties label-free selection once realistic scatter and noise enter, and is overtaken only where the physics is nonlinear.

While SpectralForge captures key primary photophysical phenomena, the simulator operates under certain physical and optical simplifications. Currently, inner-filter attenuation and reabsorp-

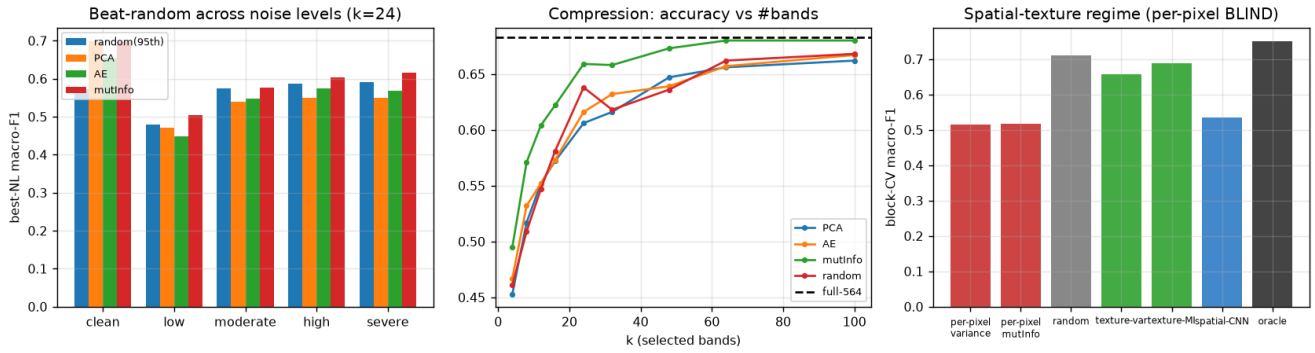


Fig. 6. The benchmark in use. Left: beat-random across five noise levels: blind PCA and autoencoder selection beat the random 95th percentile only on clean data, and under clutter only supervised mutual information does. Middle: compression curve: the autoencoder tracks PCA at every budget, both reaching about 95% of the full spectrum, while labels compress about 2.7× better. Right: the spatial-texture regime under block cross-validation: per-pixel selectors are the worst (below random) because the classes have identical per-pixel spectra, and only a spatial model with the right bands reaches the oracle.

tion are modeled using first-order exponential path approximations rather than full 3D radiative transfer or Monte Carlo photon transport through dense scattering media. Furthermore, the instrument model assumes an idealized optical train with spatially uniform illumination, uniform spectral detector response, and static photophysics, omitting dynamic exposure photobleaching kinetics during sequence acquisition as well as chromatic aberrations and thermal drift. Expanding the engine to incorporate full radiative transport and dynamic bleaching models remains an important direction for future extensions.

The same generator supports experiment planning, algorithm prototyping, and teaching. A natural next step is to evaluate performance on complex, multi-component target samples (heterogeneous biological tissues, environmental specimens, or engineered multi-fluorophore composites), where informative spectral structure is decoupled from overall band variance. SpectralForge and its workbench are in active development toward a production release, and the authors can provide exclusive early access to interested laboratories and individuals on request (please contact the corresponding author by email); the full experiment record accompanying this study is released for reproducible use.

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